Jakob Shurtleff

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EDUCATION

B.S. in Physics December 2027 - December 2027

Brigham Young University - Idaho Novi, Michigan

- emphasis on computational methods and machine learning application

EXPERIENCE

Physics Lab Coordinator & Teaching Assistant February 2026 - Present

BYU-Idaho Physics Department Rexburg, ID

- Coordinated operations and logistics for the university physics homework help center, facilitating peer instruction and academic support resources.
- Served as Teaching Assistant for Physics 121 (Principles of Physics I); clarified mechanics concepts, and graded coursework.

Volunteer Physics Tutor August 2025 - February 2026

BYU-Idaho Physics Department Rexburg, ID

- Conducted one on one tutoring to help physics students bridge the gap between complex principles and systematic methodologies for solving problems.

Door-to-Door Sales Representative April 2024 - July 2024

Glide cleaners Bountiful, UT

- Executed direct-to-consumer sales, honing negotiation, objection handling, and pipeline management in a performance-driven environment.

TECHNICAL SKILLS

- Languages & Libraries: Python (NumPy, Polars, Scikit-learn, Matplotlib, pytorch), C++, C#, SQL.
- Developer Tools: Git, Visual Studio Code, Linux, REST APIs.
- Domain Focus: Numerical simulation, machine learning deployment, process automation, data analysis.

TECHNICAL PROJECTS

AI projects

Independent Computational & AI Development

- Engineered a Physics Informed Neural Net (PINN) for modeling physical systems based on noisy data.
- Built custom LLM utilities and agential workflows, using pytorch and polars for data and training pipelines.

Nested sampling research April 2026 - Present

Research under Dr. Lance Nelson.

- Utilized HPC resources to perform large scale simulations of silicon using Nested Sampling to determine viability of silicon in outperforming current materials.

CERTIFICATIONS

- AI Engineering and Development - Coursera (2025).